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TASK 2: REQUIREMENT GATHERING

Submitted to:

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1. STAKEHOLDER IDENTIFICATION

The stakeholders are the individuals, groups, or entities that have an interest in the development, operation, or outcomes of the system. The stakeholders are categorize into primary stakeholders (directly involved) and secondary stakeholders (those with an indirect interest).

1.1. Primary Stakeholders (Directly Involved)

1.1.1. Mobile Network Operators (MNOs)

(e.g., MTN, Orange, Camtel). These are the main beneficiaries of this system. They will use the data to optimize their network, perform customer experience management, and make data-driven decisions to reduce churn and improve service quality.

1.1.2. Mobile Subscribers (End-Users)

The users who install and run the app. Their role is to provide the raw data (subjective feedback and device metrics). They benefit from potentially better network performance and the ability to voice their experiences.

1.1.3. App Development Team

These are the individuals (you which are responsible for designing, building, testing, and maintaining the mobile application and the backend infrastructure that processes the collected data.

1.2. Secondary Stakeholders (Indirect Interest)

1.2.1. Telecommunications Regulatory Authorities (e.g., TRB in Cameroon)

They have an interest in using such data to ensure that MNOs are meeting their service level agreements (SLAs) and quality of service (QoS) requirements set for the country.

1.2.2. Network Engineers and Optimization Teams

Internal staff at the MNO who will analyze the data to pinpoint specific

geographical areas with poor performance, allowing them to dispatch maintenance teams more effectively.

1.2.3. Data Privacy & Compliance Officers

Individuals responsible for ensuring that the app adheres to national and international data protection regulations, especially since it collects location and performance data from user devices.

1.2.4. Marketing and Strategy Departments

MNO departments that can use aggregated QoE data to understand customer sentiment, segment users based on their experience, and create more targeted service offerings or promotional campaigns.

2. REQUIREMENTS GATHERING TECHNIQUES

The requirement gathering phase is a critical step in the software development lifecycle. It involves identifying, collecting, and documenting the needs and expectations of all parties involved in or affected by the system being developed. For this project, which involves the design and implementation of a mobile application for collecting user experience data from mobile network subscribers in Cameroon, requirement gathering was conducted using multiple complementary techniques. These techniques include surveys, interviews, brainstorming, document analysis, and reverse engineering. The use of multiple techniques ensures that the requirements gathered are comprehensive, accurate, and reflective of real user needs.

2.1. Techniques Used

For this our project, we used three major requriments gathering techniques to get of data. The requirements are how they were used are stated below:

2.1.1. Surveys

A survey was designed and administered to mobile network subscribers as the primary requirement gathering technique. Surveys were chosen because they allow data to be

collected from a large number of people quickly and efficiently, and they are particularly effective when the target population is geographically spread across different regions.
The survey was structured into four sections covering personal information, network usage habits, network experience and problems, and attitudes toward the proposed application. Questions were deliberately written in simple, non-technical language to ensure that respondents of all backgrounds could understand and answer them comfortably. A total of 25 questions were included, combining multiple choice questions, select-all-that-apply questions, scale ratings question, and one open-ended question at the end.
The survey was distributed digitally using Google Forms and shared through WhatsApp groups and other social media platforms to reach subscribers across different regions of Cameroon including Douala, Yaoundé, and other areas. This approach allowed responses to be collected quickly while reaching a diverse group of subscribers.
The survey aimed to uncover the following:

- The most common network problems experienced by subscribers
- The times and locations where network quality is worst
- Whether subscribers use multiple SIM cards and switch between networks
- The willingness of users to install and use a background monitoring app
- The concerns and barriers that might prevent users from adopting the app
- The features users would find most valuable in such an application

The responses collected from this survey formed the primary basis for identifying both functional and non-functional requirements of the proposed system.

2.1.2. Brainstorming

Brainstorming sessions were conducted among the team to generate and evaluate ideas for the application's features, design, and technical approach. This technique was used to complement the data gathered from surveys and interviews by encouraging creative thinking about what the system should do and how it should behave.
During brainstorming, the following key questions were discussed:

- What network metrics should the app collect automatically in the background?
- How should the app prompt users for feedback without being intrusive or annoying?
- How frequently should data be collected to be useful without draining battery life?

- What should happen when the app detects that network quality has dropped significantly?
- How should collected data be stored on the device and presented to the user?

These sessions produced several important design decisions including the decision to implement a background service for continuous monitoring, the use of periodic non-intrusive notifications for user feedback prompts, and the inclusion of a personal network quality score to make technical data accessible to everyday users. Brainstorming also helped identify potential risks and challenges early in the development process, allowing the team to plan solutions in advance.

2.1.3. Reverse Engineering

Reverse engineering was used as an additional requirement gathering technique by studying existing mobile applications that perform functions similar to the proposed system. This involved downloading, testing, and carefully analyzing three widely known network monitoring applications; Speedtest by Ookla, nPerf, and OpenSignal, to understand their features, strengths, and limitations.

- Speedtest by Ookla

Speedtest is one of the most widely recognized network testing applications globally. It allows users to manually measure their download speed, upload speed, and ping. It stores a history of past test results and displays the user's current network operator. But, Speedtest is entirely user-initiated, meaning the user must open the app and actively run a test. It does not monitor network quality continuously in the background, does not collect subjective user feedback, and only captures a snapshot of network performance at the specific moment the user chooses to test. These limitations mean that it misses the everyday real-world experience of subscribers.

- nPerf

nPerf is a network testing application that collects data from user-run tests to build coverage maps showing network quality across different locations. It offers a useful geographic visualization of network performance. However, like Speedtest, nPerf operates entirely on a manual, user-initiated basis. Background test scheduling is only available in its enterprise version designed for professional field technicians and is not accessible to regular subscribers. nPerf also does not collect any subjective feedback from users about their

experience.

- OpenSignal

OpenSignal is the closest existing application to what the proposed system aims to achieve. It collects signal strength and network type data, provides operator comparison features, and has some capability for passive background data collection. It also covers Cameroon in its global database. However, OpenSignal is not widely known or used among everyday Cameroonian subscribers, and its interface is relatively complex for non-technical users, and it does not incorporate any mechanism for collecting subjective user experience feedback or allowing users to describe their experience in their own words.

Studying these three applications revealed a consistent and significant gap in existing tools. None of the existing applications combine continuous background monitoring with subjective user feedback collection in a single tool designed for everyday subscribers.

To conclude, this analysis confirmed that there is a clear need for a solution that runs continuously in the background, collects both objective network metrics and subjective user ratings, and is designed with the specific context of Cameroonian mobile subscribers in mind. These findings directly informed the functional requirements of the proposed system

The combination of surveys, interviews, brainstorming, document analysis, and reverse engineering provided a comprehensive and well-rounded foundation for identifying the requirements of the proposed system. Each technique contributed a different dimension of understanding — surveys provided broad quantitative data from many users, interviews provided deeper qualitative insight, brainstorming generated and evaluated design ideas, and reverse engineering identified gaps in existing solutions that the proposed system must address. Together, these techniques ensured that the requirements gathered are grounded in real user needs, technically informed, and clearly justified.

3. Data Gathering

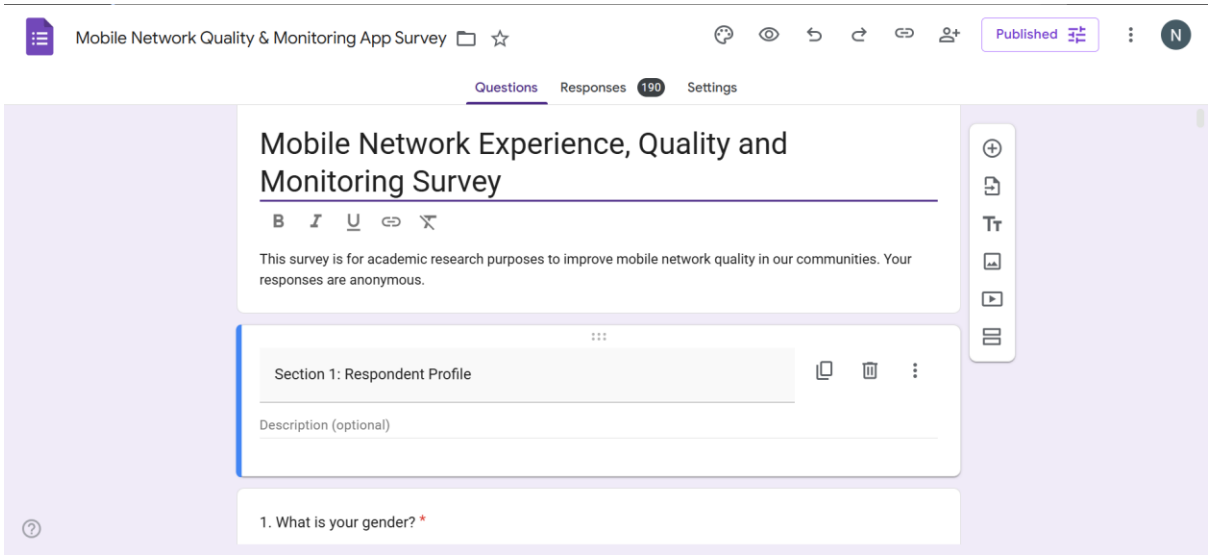
Data gathering is the systematic process of collecting, measuring, and analyzing accurate information from various sources to answer research questions, solve problems, or evaluate outcomes. In the development of our mobile applications for this project, the data gathering phase is crucial for bridging the gap between technical network performance and the actual user perception (Quality of Experience).

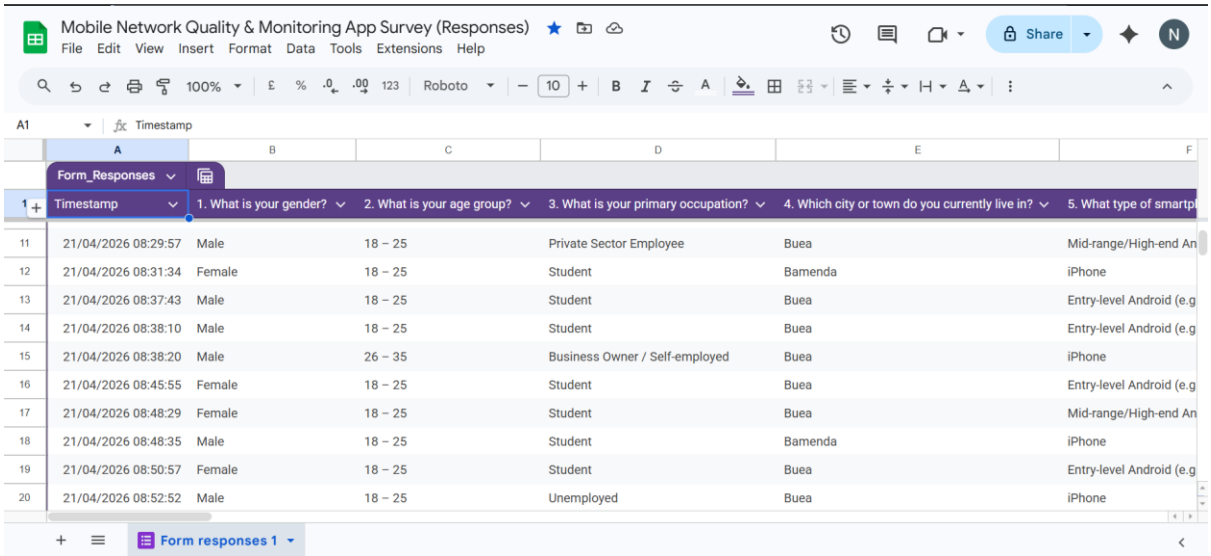
3.1. Key Data Gathering Methods

- **Surveys/Questionnaires:** Often used for collecting quantitative data from large groups.
- **Interviews:** One-on-one or group discussions that provide in-depth, qualitative, or quantitative data.
- **Observations:** Recording behaviors or events directly.
- **Data Mining/Secondary Data:** Utilizing existing data sources like government databases, reports, or research studies.
- **Focus Groups:** Small, guided groups discussing specific topics for detailed perspectives.
- **Experiments/Testing:** Performance measures used to evaluate outcomes.

3.2. Data Gathering Process

- **Identifying Objectives:** Here, we determined what data is needed and why. In this phase we came out with all the necessary questions for requirements gathering phase that will lead to a success.
- **Selecting Methods:** Here, we chose the appropriate tools (e.g., google form surveys, reverse engineering and brainstorming) which were based on goals of this project.
- **Preparing & Pilot Test:** We then went ahead to create a google form for our data collection and also created a questionnaire to ensure reliability in what we were doing.
- **Collecting Data:** After successfully created our google form and connecting it to an excel sheet, we went ahead to sharing the google form to different people of different age groups to answer the form about what the think. We executed it following all ethical standards. A sample of our survey form and excel sheet is shown below;





- **Analyse & Interpret:** After we had reached a value of 188 respondents to our form, we went ahead to analyse our results and more details will be given in the next task (Requirement analysis).

3.3. Key Considerations

- **Ethical Concerns:** We ensured participant confidentiality and informed consent. We did this by making the answers to the form anonymous where the respondent was not expected to fill in their email or any other personal information.
- **Quality:** We ensured accurate data collection for crucial research integrity. We did this by letting all the different corresponds to give us feedback on what the think.
- **Type:** We used qualitative (descriptive) data based on objectives to collect our

data.

4. Data Cleaning

It is the process of identifying and correcting or removing inconsistencies, inaccurate, incomplete or irrelevant records from raw data sets to improve data quality.

4.1. What is the aim of Data Cleaning?

The goal of data cleaning is to help ensure that data is accurate, complete, consistent and usable for analysis or decision-making. Data cleaning processes work to address common data quality issues such as duplicates, missing values, inconsistencies, syntax errors, irrelevant data and structural errors. By fixing these errors, it ensures that analytics, machine learning models, and business decisions are based on accurate, reliable, and consistent information, ultimately improving efficiency.

4.2. Process of Data Cleaning

There are five major steps involved in the process of data cleaning. They are listed below:

- **Remove duplicate or irrelevant observations**

  Duplicate data is a common issue that can be tackled first. It is often the result of system errors, manual data entry mistakes or the integration of multiple datasets. Identifying any instances of repeated or irrelevant data and removing it makes the dataset more efficient and accurate.

- **Fix structural errors**

  Different data collection tools may use different classes or data formats, while manual data entry often leads to typos or mistakes like inconsistent capitalization. These errors can cause structural compatibility issues downstream.

- **Eliminate data noise**

  Identify outliers and noise (irrelevant data points which should be ignored during analysis) and remove them from the dataset to prevent them from negatively impacting the decision-making process.

- **Handle missing data**

  Connectivity issues, hardware failures or other problems can lead to missing points in

the dataset, which can reduce its usability and value. Addressing these gaps by recreating the data or adjusting the range of the dataset can help maintain its continuity and value.

- **Validate and verify**

  Validation is the final step in the cleaning process, and it can involve both manual and automated checks to ensure that each of the above potential issues has been caught and addressed.

Our data was cleaned following the data cleaning processes mentioned above.

4.3. Process of Cleaning our Data

i. **Structural Standardization:** The original column headers were full-length survey questions, which often cause errors in data tools.
   - Renaming: Every column was shortened to a precise word which still defines thesame thing (e.g., "What is your primary occupation?" became ‘occupation’). And we did this to all the column heads of our data.
   - Whitespace Removal: We performed a "Trim" operation on every text cell to remove accidental leading or trailing spaces that could cause duplicate categories (like "MTN" vs "MTN ").

ii. **Geographic & Text Harmonization:** This was the most significant area of cleaning. Raw text input often leads to inconsistent naming.
   - City Name Consolidation**:** We resolved over 15 variations of city names.
     - Case Correction: Converted "buea" and "BUEA" to "Buea".
     - Accent Normalization: Standardized all "Yaoundé" to "Yaounde" for better searchability.
     - Neighbourhood Mapping: Mapped specific locations like "Molyko" and "Bomaka" to their parent city, "Buea".
     - Typo Correction: Fixed entries like "Buae" to "Buea".
   - Network Labels: Entries for "All of the above" were expanded to list the specific providers ("MTN, Orange, Camtel") to ensure they are captured when filtering for specific networks.

iii. **Data Integrity & Deduplication**: To ensure the statistical results are not skewed by repeat entries:

- Duplicate Removal: We identified and deleted 1 duplicate row where a respondent had submitted the form twice with identical answers.
- Missing Value Management: In the additional_comments field, empty cells were replaced with "No comment" to ensure the dataset has no "null" values, which can break some Excel formulas.

iv. Data Reduction (Filtering)

- Column Removal: As per your specific request, we removed some of the data with less relevance for now for the aim of the project. The following columns; 'what is your gender', 'Do you use more than one SIM card (Dual sim)?', 'Have you ever missed something important (exam, work deadline, or payment) due to poor network?' and 'Have you ever formally complained to your operator about these issues?' were completely removed to focus your analysis on the remaining metrics.

After performing all the above operations, our data considered clean and we'll use the data for our data analysis.

5. User Reluctance Assessment

User reluctance assessment is a critical, often overlooked, component of requirements engineering, focusing on identifying why stakeholders may resist adopting new systems or processes. It aims to identify potential bottlenecks—such as lack of trust, fear of change, or increased cognitive load—early in the development lifecycle to ensure successful adoption.

5.1. Core Factors Driving User Reluctance

Analysis indicates that reluctance often stems from:

- **Perceived Lack of Value:** Users do not see the benefit of the new technology.
- **Lack of Confidence/Skills:** Users feel overwhelmed or lack the competence to use the new system.
- **Workflow Disruption:** New tools interfere with established, efficient working methods.
- **Trust and Security Issues:** Users fear data loss, reduced autonomy, or surveillance.

- **"Invisible" Workloads:** Technology may add hidden, time-consuming steps, such as data entry or managing automated systems.

A major challenge faced during the early stages of this project was low user participation and engagement in data collection activities.

5.2. Key Issues:

- Many of the correspondents complained about the number of questions in the survey saying the questions were too many (25).
- Some people could not access the google form because of network issues where the form could not load in their phones.
- Many users were unwilling to fill out survey by ignoring it when it was sent to them.
- Some people were scared of spam and hack saying they are afraid the form can extract information from them that can be used against them and others were afraid their accounts may be hacked if the fill the form so the needed confirmation from us to make sure the site was safe.
- Most older people (50yrs and above) and above didn't really collaborate as some claim not to have time and others who didn't even know what to do with the form.
- Low response rates from distributed Google Form.
- Some participants ignored or abandoned surveys midway.
- Difficulty accessing IT professionals due to lack of permissions or formal authorization.
- Limited opportunities and time to conduct in-person interviews in professional environments.

5.3. Implications:

- Limited primary data for decision-making.
- Potential bias due to small or non-representative sample sizes.
- Increased reliance on assumptions or secondary research

5.4. Insights

- Users are generally hesitant to invest time in feedback processes.
- Lack of trust or perceived value in participation.
- Formal access barriers when gathering expert-level insights.

5.5. Recommended Mitigations

- Simplify surveys to reduce completion time.
- Provide incentives or clearly communicate the value of participation.
- Use more interactive data collection methods.
- Seek institutional or academic authorization when approaching professionals.
- Combine multiple data collection strategies.

Conclusion

In conclusion, effective requirements gathering is the critical foundation for project success, bridging user needs with technical solutions to ensure stakeholder alignment, minimize scope creep, and reduce costly rework. It defines clear project objectives, sets realistic expectations, and provides a structured roadmap for development. A thorough, documented, and validated set of requirements is crucial to deliver value, ensure quality, and keep projects within budget.

Key Takeaways

- **Foundation for Success:** Clear requirements are the most important factor in preventing project failure and reducing rework.
- **Stakeholder Alignment:** It ensures all parties share a common understanding of project scope and objectives.
- **Risk Management:** Early identification of requirements allows for better risk assessment and mitigation.
- **Validated Documentation:** A well-documented, approved, and prioritized set of requirements serves as a reliable roadmap.
- **Continuous Engagement:** Successful gathering is a collaborative, ongoing process, not a one-time event.

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